

1. Childhood Obesity Prevention Study

Data Introduction

You have data from 300 children enrolled in a community-based obesity prevention program. Each child is weighed at baseline (age 8) and then annually for 4 years (ages 9, 10, 11, 12). The main outcome is **BMI percentile**. Key measured factors include:

- **Physical activity** (minutes per day, self-reported)
- **Dietary habits** (measured using a food frequency questionnaire, e.g., daily sugar-sweetened beverage intake)
- **Socioeconomic status** (SES), categorized as low, medium, or high
- **Gender** (male/female)

Hypothesis of Interest

Whether the **BMI percentile** trajectory over time differs based on participation in the program and how these differences are moderated by **SES** or **gender**.

2. Blood Pressure in a Multi-Center Clinical Trial

Data Introduction

A pharmaceutical company sponsors a new **antihypertensive medication** trial. The study recruits 500 patients from 10 different clinics. Each patient's **systolic blood pressure (SBP)** is measured at months 0, 1, 3, 6, and 12. Each patient is randomized to **Drug A** vs. **Placebo**.

Additional information:

- **Age** (continuous)
- **Clinic ID** (to capture site-level differences)
- **Medication adherence** (percentage of prescribed doses taken)
- **Baseline comorbidities** (e.g., diabetes: yes/no)

Hypothesis of Interest

Patients on **Drug A** will have a **steeper reduction** in systolic blood pressure over 12 months compared to placebo, controlling for clinic-level differences.

3. Environmental Exposure and Respiratory Symptoms

Data Introduction

You have daily measurements of **air pollution (PM2.5)** from 20 different neighborhoods. In each neighborhood, you have **daily counts of emergency room visits** for **asthma-related** complaints over a one-year period (365 days).

Additional contextual variables include:

- **Neighborhood average SES** (continuous, standardized)
- **Season** (categorical, e.g., winter, spring, summer, fall)
- **Day of week** (to account for systematic variation)

Hypothesis of Interest

Higher **PM2.5 levels** are associated with **increased asthma-related emergency visits**, possibly differing by **season**.

4. Mental Health Assessment in a Telehealth Intervention

Data Introduction

A mental health support program enrolls 200 individuals who exhibit symptoms of moderate depression. They receive **weekly telehealth counseling** sessions for 10 weeks. Each week, participants complete a **depression symptom score**(e.g., PHQ-9). They also report:

- **Stress level** on a 1–10 scale
- **Medication use** (yes/no)
- **Number of hours of sleep** in the past 24 hours

Hypothesis of Interest

Whether **depression symptoms** improve more quickly among those with **consistent medication use** and/or higher **sleep**quality, and whether stress level modifies the rate of change in symptoms.

5. Vaccination Coverage in School Districts Over Multiple Years

Data Introduction

You have an annual survey of **vaccination rates** for measles in 50 school districts over 5 years. For each district-year, you record:

- **Measles vaccination coverage** (as a percentage of eligible children)
- **Population density** (continuous)
- **Median household income** (continuous)
- **Geographic region** (e.g., rural vs. urban)

Additionally, a statewide policy to **increase vaccination awareness** was implemented at the start of year 3.

Hypothesis of Interest

Vaccination coverage will **increase** following the statewide policy, and the **effectiveness** of the policy depends on **population density** and **income** levels.

6. HIV Viral Load Monitoring in an Infectious Disease Clinic

Data Introduction

In an observational study, 100 HIV-positive patients are monitored for 2 years, with **viral load** measurements taken every 6 months (baseline, 6m, 12m, 18m, 24m). Data collected includes:

- **Medication regimen** (two categories: standard vs. new regimen)
- **Patient age** at baseline
- **CD4 count** at baseline
- **Adherence** (continuous measure of proportion of doses taken)
- **Time since diagnosis** (years)

Hypothesis of Interest

Patients on the **new regimen** have a **faster decline** in viral load compared to those on the standard regimen, and the effect is possibly influenced by **adherence** levels.